Project Identification and Evaluation: a Case on Business Process Reengineering of Ports Project Workflow Management

Alexander A Hernandez
School of Industrial Engineering and Engineering Management, Mapua University
Philippines
aahernandez@mymail.mapua.edu.ph

Grace Lorraine D Intal*
School of Industrial Engineering and Engineering Management, Mapua University
Philippines
gldintal@mapua.edu.ph

Abstract

Ports project management is crucial in economic development of archipelagic country in Southeast Asia to bridge mobilization of human and transportation activities. As such, a developing country such as the Philippines needs well-defined business processes to support ports project identification and evaluation. This paper aims to present a ports project identification and evaluation applying business process reengineering tools, which aims to enhance the efficiency and effectiveness of ports project identification and evaluation processes to its stakeholders. The paper presents the problems identified and analysis, the proposed solutions, and evaluation to aid decision-making and actions towards attainable ports project management. Practical recommendations are offered to further the proposed system design and improvement.

CCS Concepts

• **Applied Computing**; • **Enterprise Computing**; • **Business Process Management**; • **Cross-organizational business processes**;

Keywords

Business Process Reengineering, Systems Engineering, Project Management, Blockchain, Analytics

ACM Reference Format:
Alexander A Hernandez and Grace Lorraine D Intal. 2024. Project Identification and Evaluation: a Case on Business Process Reengineering of Ports Project Workflow Management. In *2024 the 7th International Conference on Information Management and Management Science (IMMS 2024), August 23–25, 2024, Beijing, China.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3695652.3695668

1 Introduction

Ports project management plays a vital role in economic development and mobility of archipelagic situated countries in Southeast Asia such as Indonesia, Malaysia, and the Philippines. More recently, project management has become relevant in ports agencies to sustain the development of ports strategically located in different parts of a country [1]. Besides, project management introduces avenues for ports agencies to revisit existing business processes that support the achievement of milestones and outcomes [2, 3]. In recent study, Roh et al. [4] found that efficient and effective ports project management is realized by the availability of resources such as data, tools, guidelines, and business processes that enable a wholistic approach towards project management needs.

Business processes reengineering (BPR) is crucial to ports agencies to address inefficiencies concerning project management such as on ports identification and evaluation [5]. BPR aims to simplify and eliminate redundant and complex processes affecting customer satisfaction and the organization's performance [6]. Besides, BPR supports change management and performance measurement through cross functional collaboration, technology enablement, and radical design techniques [7]. Further, BPR is a process focused with customer centricity which may contribute to effective and efficient delivery of project management milestones and outcomes [8].

However, several ports agencies in Southeast Asia suffer delays on ports project development and management [9, 10]. This situation affects regional, national, and cross-boundary integrations to support trade and economic development [11]. Further, ports agencies in Southeast Asia experience lacks technology enablement to support ports project identification and management resulting in project failures and unstable conditions of existing ports [12, 13]. As such, BPR can potentially address these growing issues and concerns to assist various levels of stakeholders from operational, managerial and decision-makers to remain effective and productive in meeting the demands of ports project [14]. Similarly, BPR as applied to ports project identification and evaluation can assist port agency to reduce cost while optimizing processes undertaken which can radically enhance ports environment [15].

Recent advances on blockchain and analytics have been evolving in ports projects and development initiatives [16]. Blockchain enables ports to integrate supply chain flows concerning its physical, information and financial flows while involving various parties involved to share certain information in efficient and transparent approaches [17]. Further, blockchain provides potential to document, validate, and secure each event in the chain [18]. Besides, blockchain technologies are complemented with analytics for operational and decision-making activities to make coordinating of roles and access to information efficient from monitoring of ports to linking supply chain processes that brings end-to-end value to all stakeholders. [19, 20].

In this paper, a case study on ports agency is undertaken to apply business process reengineering aiming to address inefficiencies

*corresponding author.

IMMS 2024, August 23–25, 2024, Beijing, China

ACM ISBN 979-8-4007-1699-7/24/08
https://doi.org/10.1145/3695652.3695668

Table 1: Mckinsey's 7s Framework of the Ports Agency

Strategy	Enhance accessibility through seamless connectivity with other transport modes at par with global best practices and good governance.
Structure	Hierarchical - divisional structure
Systems	Maritime facilities, financial resource, intellectual resource (internal experts)
Shared Values	Transparency, Reliability, Innovation, Professionalism, Sustainability, Social Responsibility
Style	Collaborative, customer-focused culture.
Staff	Employees are well educated, centralized - not autonomous.
Skill	Project management, engineering and design, information technology competencies,

in ports project identification and evaluation. The current ports agency project identification and evaluation is assessed and evaluated leading to the proposed system design based on workflow management system integrating blockchain and analytics.

2 Methods

This section describes the case study, environmental scanning, the SWOT analysis, and current processes with problems identified, which are aimed to support the proposed system design and improvement. This study is based on business process reengineering that proposes enhancement of ports project identification and evaluation. To proceed with BPR, this study conducted process reviews, observations, and interviews with stakeholders to collect relevant information to elicit the problems and challenges present in the existing systems. After such, the data gathered were used for further analysis, which is supported by several presentations to the stakeholders to review, and confirm information provided. Some suggestions were taken during the consultations to assist in proposing enhanced processes aiming to improve efficiency and productivity of end users. While the case study organization is based on local context, the proposed enhancements were based on globally accepted standards and techniques to ensure compliance to regulatory requirements and policies.

2.1 The Case Organization

This paper presents a ports agency as a case organization in the project identification and evaluation. The case organization aims to provide integrated transport and logistics systems. Besides, the case organization establishes, develops, regulates, manages, and

Table 2: SWOT Analysis - Ports Project Identification and Evaluation - SWOT Analysis

Strengths •Strategic development of local and international servicing ports •Ports project development to accommodate 4th generation vessels. •Using emerging technologies to support project related procurement and bidding activities. •Government support for land use. •Geographical locations.	**Weaknesses** •Availability and timeliness of ports operations data from the local government (e.g., ship, cargo, & passengers' statistics -reporting monthly deduction, port management office - months / years no data). •Extensive bureaucracy, political stability, commitment of LGUs to regional integration / checkpoints in local government units (e.g., delays in approval of the detailed engineering port design). •Lacking information technology / systems improvement to support ports project feasibility studies (e.g. leading to inaccurate estimation of project identification / due to human error, excel file referencing from other files are prone to error). •Limited port capacities •Identification of trade lanes positions locally and Southeast Asia •Poor research practices, lack of collection and analysis of port related data (e.g., no guidance of port subject matter expert on the use of existing forecasting method to be used)
Opportunities •Potentially strong growth anticipated in new port hubs for transshipment and tourism. •Regional integration efforts between private and public ports •Rehabilitation / upgrade of existing ports. •New energy projects to support ports development and operations. •Public-private partnerships on new projects.	**Threats** •Slow procurement of governmental permissions. (e.g., LGU participation in project identification - issues on illegal settlers for the potential project site / development). •Recent security attacks on regional ports and terminals. •Poor conditions related to criteria for design parameters (e.g. such as the zone of significant port interest containing the requirements - road networks, adjacent ports) in some provinces. •Competition from regional ports

operates a rationalized national port system in support of economic development.

This project briefly presents the Mckinsey's 7s Framework to summarize the strategy, structure, systems, shared values, style, staff, and skills applied in the case organization, as presented in Table 1.

2.2 Environmental Scanning and Analysis

This project aims to focus on the Engineering Division aiming to address the inefficiency challenges in project identification and evaluation. To broaden understanding of the ports, this project presents the SWOT analysis (strengths, weakness, opportunities, and threats), ass presented in Table 2.

2.3 Fishbone Diagram of Project Identification and Evaluation

This project presents the cause and effect (Ishikawa) diagram to provide an overview of the root cause of the inefficiency in project identification and evaluation in the Engineering division, as presented in Figure 1.

2.4 Current Processes on Project Identification and Evaluation

Figure 2. presents the current processes on project identification and evaluation covering major steps such as initiation, feasibility, analysis, evaluation, approval, and identification close out. Further, this paper presents the problems in ports project identification and evaluation, ass presented in Table 3.

3 Results and Discussion

This section presents the decision analysis, cost-benefit analysis, and failure mode and effect analysis when the proposed system design and improvement is implemented.

3.1 Decision Analysis

Table 4. shows the alternative options for the workflow management system which can be considered in the selection of the most suitable solution for current problems in project identification and evaluation. After consultation with the end-users and stakeholders involved in the case study, the group decided to provide weights for each criterion identified necessary in the selection of the best alternative option for the given problem. The criteria include usability, functionality, efficiency, auditability, interoperability, and security.

The first alternative is a project management software which primarily focuses on providing a workflow of specific project milestones and provision for the deliverables. The second alternative is the blockchain-based WMS with analytics that utilizes state of the approach to manage document workflows while providing analytics dashboard to support actionable insights on the project identification and evaluation progress and outcomes. The AI driven workflow automation primarily concentrates on diffusion of algorithms that reduces human intervention on document workflow processes needed to manage project identification and evaluation. Furthermore, AI based workflow automation is fully reliant on providing a pipeline of the necessary inputs, checkpoints, business logic, and outputs to implement the artificial intelligence components to attain its full automation functionality. As a result of assessment and evaluation, the most appropriate alternative is the

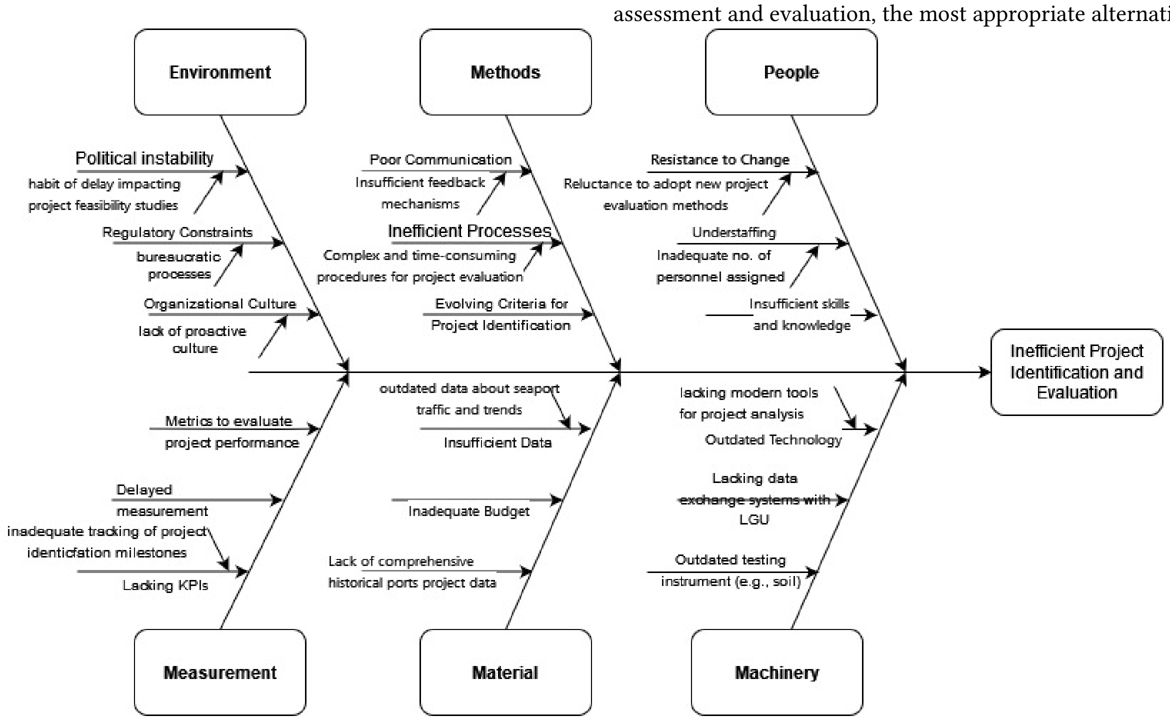

Figure 1: Cause and Effect Diagram on Inefficient Project Identification and Evaluation

Table 3: Project Identification and Evaluation Description and Problems Identified

Steps	Description	Problems Identified
1	**Initiation** - this step requires identifying the need or opportunity based on historical data of outports such as cargo shipment, passenger traffic, and other utilization reports of existing outports in the country. Each outport is expected to submit monthly ports traffic and trends reports. The increasing traffic on ports is one of the bases (variables) / requirements for initiation, conducted by the Engineering Division. The initiation also covers stakeholders affected such as local government units where outports are situated, residential communities, and environmental groups potentially affected. This step defines whether the initiation is for outport upgrade (e.g., extending the capacity to increase utilization), maintenance of existing outports, or proposals for new ports in the country. This step results in the initiation report needed to proceed with feasibility study.	Data reporting timeliness and availability, traffic forecasting for port identification and planning, and historical project development.
2	**Feasibility** - this step needs the parameters such as annual port statistics, gross domestic product / gross state product, zone of significant port interest, and forecasting, among others to develop the feasibility study assessing its technical, economic operational and scheduling, and regulatory aspects of the potential project identified for an existing output or new port in the local government unit. The potential risks and constraints are also identified (such as regional integration, competition, land use etc.), that may impact the project's success. Further, developing estimated budget and resources through the assistance of the Detailed Engineering Division, needed to execute the project.	Coordination with internal and external stakeholders, inadequate tracking of project identification milestones and objectives.
3	**Analysis** - this step requires the analysis made on the technical, financial, market, and regulatory feasibility aspects of the project identified. Each project identified is subjected to presentations of the Engineering Division to the Evaluation Committee to properly assess the needs for upgrade, maintenance, or new outports to support cargo shipment and passenger services.	Inadequate qualified personnel to perform the analysis leading to the inefficiency of the national port network and imbalance in port investments.
4	**Evaluation** - this step covers risk assessment, cost-benefit analysis, social and environmental impact assessment of the potential project identified. Whenever necessary, the local government unit and affected communities are invited during the evaluation process to provide useful comments and suggestions to improve the project identified details. The iterative revisions happen in this phase to conform with all applicable requirements and conditions set by all parties such as port agency, local government, and other affected communities of the seaport project identified. The evaluation committee prepares and submits the evaluation report with annotations for the approval of the port agency board.	Feedback loops delay the evaluation of the project identified, Insufficient metrics to evaluate project identified (e.g., international standards and benchmarks of evolving requirements).
5	**Approval** - this step requires an analysis and evaluation report of the project identified with annotations / changes. The port agency board convenes to present and discuss the project identified, as well as possibly secure funding (e.g., government grants, loans, private investment, and public-private partnerships) for the project identified. Necessary approval of government permissions will be prepared before the identification closes out.	Disclosure of all information and actions taken involving internal and external stakeholders in all steps undertaken.
6	**Identification Close Out** - this step proceeds with finalizing the project identified scope, mobilization of resources, monitoring and control plans, development of the project plan, and project identification close out.	

blockchain-based WMS with analytics which features do not eliminate human knowledge and expertise in the process while ensuring secure, transparent, and efficient delivery of project identification and evaluation milestones.

3.2 System Design and Improvement

Figure 3. presents a comparison of existing versus enhanced workflow management system that presents fundamental components such as user experience, logging and auditing, document versioning, and interoperability. The enhancement aims to answer the complexities of the conventional workflow management by incorporating blockchain and analytics. While a review of workflow management is considered, the paper proposes an enhancement to aims to enhance user experience through mobile and web-based systems, workflow approval, logging and auditing, and versioning through blockchain technology. The enhancement also introduces interoperability features to support transparent data sharing and governance, and integration with existing or future systemsAll transactions are logged to build analytics of the current progression of the project identification and evaluation steps. In this way, all stakeholders can act about the work processes needing endorsement and approvals.

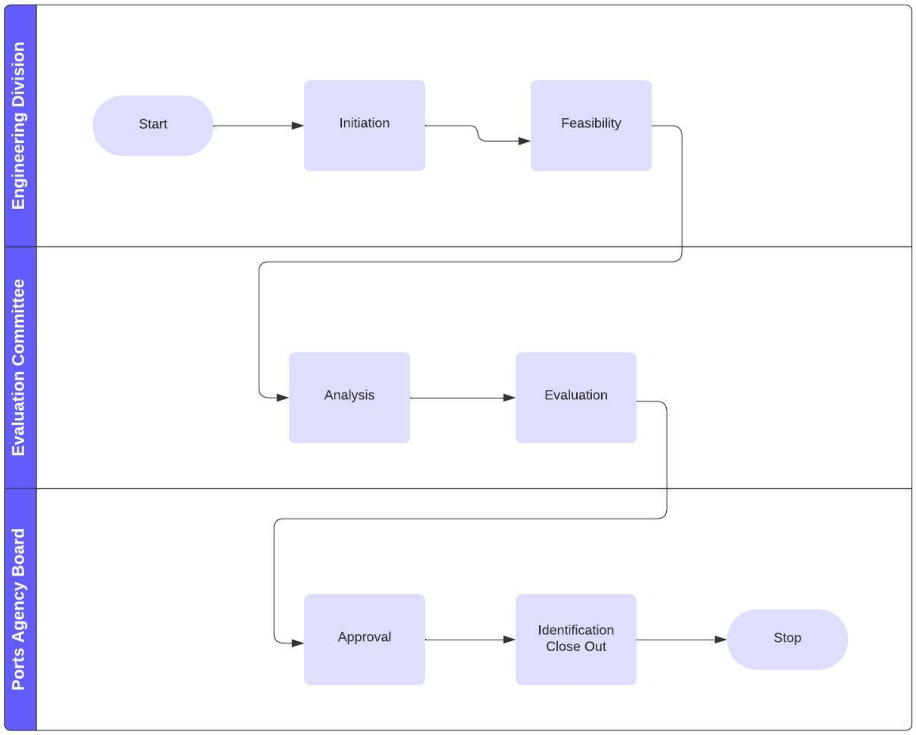

Figure 2: Project Identification and Evaluation

Table 4: Weighted Scoring of Alternative Options for Workflow Management System

Criteria	Weight	Alternative(Option 1)Project Management Software Score	Alternative(Option 2)Blockchain-based WMS with Analytics Score	Alternative(Option 3)Artificial Intelligence Driven Workflow Automation Score
Criteria 1: (Usability)	10%	8	7	6
Criteria 2 (Functionality)	20%	16	18	18
Criteria 3: (Efficiency)	20%	16	17	18
Criteria 4: Auditability	10%	8	8	6
Criteria 5: Interoperability	20%	16	18	17
Criteria 6: Security	20%	15	18	16
	Total	79	86	81

As such, the analytics can help all end users and stakeholders are informed of all progress outputs vital to decision-making activities. Similarly, the analytics supports efficiency of other departments involved reducing delays of acquisition of information and future procurement activities related to the project.

Figure 4. presents the enhanced project identification workflow by incorporating blockchain through the major steps to create a distributed leger that aims to provide transparency on the actions taken of all actors in the workflow. The document integrity layer embedded in the enhanced system aims to account for each alteration, added or deleted to represent what is exactly changed by the actors of the ports agency in the project identification workflow.

The proposed enhancement considers some real-life assumptions to ensure that the project is feasible, secure and valuable to the stakeholders involved. This study considered the technological infrastructure within and outside the port's agency, positive

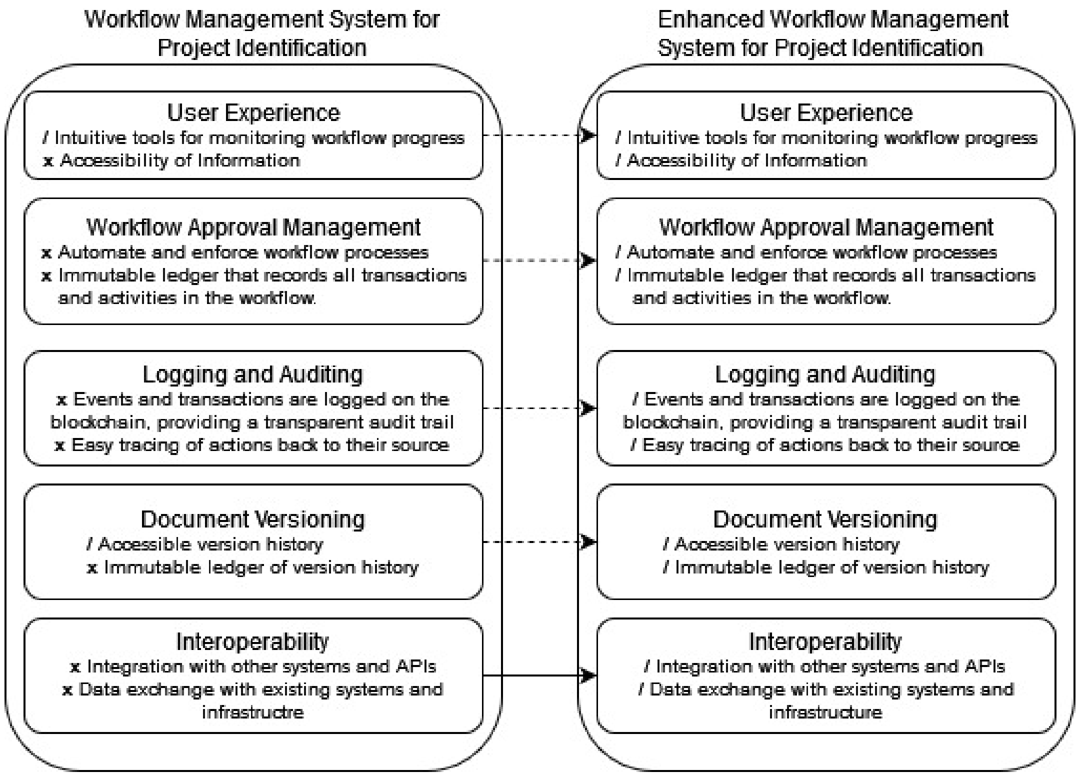

Figure 3: Existing vs Enhanced Workflow Management System for Project Identification

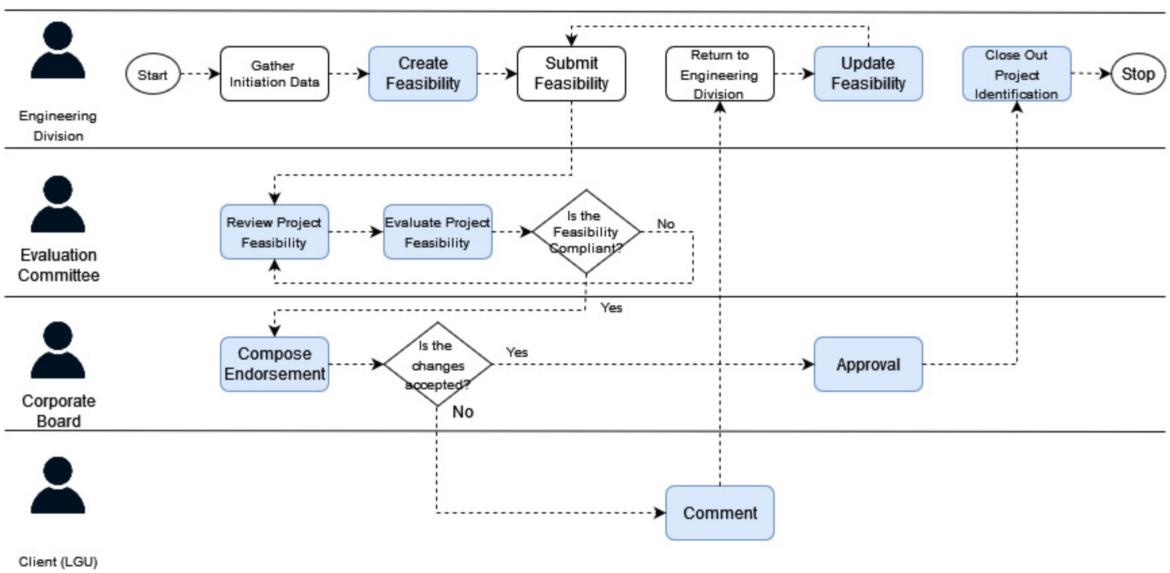

Figure 4: Project Identification Workflow Management System with Blockchain and Analytics

Table 5: Cost-Benefit Analysis of Blockchain based WMS with Analytics

Cost	Amount	Benefits	Amount
Development and Configuration Cost (1st year) •Software development •Cloud Services •Blockchain Network	 3,000,000.00 300,000.00 1,000,000.00	Transparency and Traceability (savings from reduced disagreements / disputes)	4,000,000.00
Training and Education (1st – 3rd year) •Training cost •User Manual	 150,000.00 50,000.00	Functional Performance and Efficiency (workflow automation and process efficiency)	5,000,000.00
Deployment and Maintenance (1st – 3rd year) •Deployment cost •Maintenance cost	 100,000.00 50,000.00	Auditability and Compliance (savings from regulatory requirements and enhanced auditing)	3,000,000.00
End-User Support •Technical support	 100,000.00	Analytical Insights (savings from gaining actionable insights related to project identification)	5,000,000.00
Total Cost	4,750,000.00	Total Benefit	17,000,000.00

stakeholder engagement, the adoption of data integrity and security ensuring ports project identification and evaluation accurate, complete, and up to date. Likewise, the proposed enhancement is compliant with existing local and international regulations and frameworks while ensure auditability to satisfy its regulatory requirements. With regard to the blockchain solution, performance and scalability is another assumption ensuring reliable transaction speed without compromising performance when handling large information from end users and projects. Similarly, the ports agency is assumed to have enough budget to invest on the proposed enhancement, its implementation and maintenance costs. Furthermore, the environmental impact of using the proposed system will target reduced energy consumption, making it environmentally sustainable following some green alternatives in the selection and procurement of related systems and devices needed for the project identification and evaluation.

3.3 Cost-Benefit Analysis

Table 5. presents the cost-benefit analysis of the best solution considered in the given problem. The cost-benefit analysis focuses on the major item cost needed in the development, implementation, and maintenance of the blockchain-based WMS with analytics. Cost items cover development and configuration, training and education, deployment and maintenance, and end user support. The development and configure and training education costs are one time cost needed for the project. While the deployment and maintenance and end user support must be paid once a year. The total cost is estimated at 4,750,000.00 on its 1st year development and implementation.

When it comes to benefits, this project expects that transparency and traceability, functional performance and efficiency gains, auditability and compliance to regulatory standards, and practical insights and actions could be realized on its 1st year resulting in a total benefit of 17,000,000.00. By comparing between the cost and benefits, the ports agency can realize that potential solution is financially viable while it provides positive return on investment to the organization.

3.4 Failure Mode and Effect Analysis

Table 6. describes the failure mode and effect analysis of the proposed blockchain-based WMS with analytics needed for the ports project identification and evaluation. Each step undertaken considers the failure mode, effect, severity, and recommended actions to undertake to solve potential problems. Furthermore, the proposed FMEA analysis can help software development teams, managers, and higher officers to anticipate possible expectations resulting from the steps identified. As such, these stakeholders could reduce chances of these failure events to happen, integrate possible strategies at the time of project planning stage of the blockchain-based WMS with analytics. As a result, end-users can realize the benefits rather than the potential issues in the project.

This project aims to present a business process reengineering case study of a port's agency which aims to address the problems identified in ports project identification and evaluation. This paper presents practical recommendations, to the ports agency, to consider a system design and improvement with technological integration. Practical recommendations could include the development of the blockchain based workflow management system with analytics, its functional modules based on user requirements taken from interviews and process reviews. It is also important to note that this project is focused on ports agencies to support e-governance. As such, a user centered design approach applying human values is necessary to develop an effective, secured, and transparent ports identification and evaluation systems. Lastly, this work provides guidance to project management team and decision-makers with inputs to support the realization of digitally enabled ports project management.

In conclusion, this paper brings avenues of ports project identification and evaluation enhancement through business process reengineering. Thus, there is a potential to increase efficiency and

Table 6: Failure Mode and Effect Analysis

Steps	Failure Mode	Effect	Severity	Recommended Action
Development and Configuration	Insufficient requirements gathering	Did not meet all end-users needs / requirements leading to possible inefficiencies and dissatisfied end-user	Medium	Comprehensive requirements gathering and analysis to meet end-users needs
	Poor Integration with existing systems	Incompatibility issues may surface leading to data inconsistencies and workflow interruptions	Medium	Comprehensive integration testing, assessment, and evaluation to resolve incompatibility problems
	Blockchain network operations and congestions	Large document workflow transaction records or network related congestions, which potentially impacts systems performance	Medium	Constant monitoring of blockchain traffic and implement scalable blockchain network to mitigate potential congestion
Training and Education	Insufficient training resources	Learning rate / performance is not attained which could potentially result to low productivity and utilization rate.	Medium	Provide sufficient training and support through end-user trainings and workshops
	Low user Engagement	Senior users may resist on the utilization of the blockchain-based WMS with analytics, which could lead to low enthusiasm.	Medium	A change management plan and strategy could be introduced to promote uptake and realization of its benefits
Deployment and Maintenace	Delayed release of software updates	Some vulnerabilities and security issues my arise from new threats, which could lead to security breaches and failures	High	Implement a regular software maintenance and deployment of new release versions, and patches
Analytics and Insights	Inaccurate implementation of business logic, algorithms, or biased analytics	Inaccurate project identification and evaluation analysis which could lead to flawed insights and potential errors in decision-making impacting the project deliverables and results.	High	Apply comprehensive algorithm implementation validation and quality checks to ensure accuracy and balanced analytical outcomes
End-User Support	Insufficient end-user support	End-users may experience downtimes and disappointment when technical issues are encountered which may impact their satisfaction and productivity	Medium	Provide comprehensive end-user support services to address technical issues and concerns when blockchain-based WMS with analytics is implemented

effectiveness of ports project identification and evaluation based on workflow management system with blockchain and analytics.

References

[1] Locatelli, G., Ika, L., Drouin, N., Müller, R., Huemann, M., Söderlund, J., ... & Clegg, S. (2023). A Manifesto for project management research. *European Management Review*, *20*(1), 3-17.

[2] Orieno, O. H., Ndubuisi, N. L., Eyo-Udo, N. L., Ilojianya, V. I., & Biu, P. W. (2024). Sustainability in project management: A comprehensive review. *World Journal of Advanced Research and Reviews*, *21*(1), 656-677.

[3] Stanitsas, M., & Kirytopoulos, K. (2023). Investigating the significance of sustainability indicators for promoting sustainable construction project management. *International Journal of Construction Management*, *23*(3), 434-448.

[4] Roh, S., Thai, V. V., Jang, H., & Yeo, G. T. (2023). The best practices of port sustainable development: a case study in Korea. *Maritime Policy & Management*, *50*(2), 254-280.

[5] Battilani, C., Galli, G., Arecco, S., Casarino, B., Granero, A., Lavagna, K., ... & Damiani, L. (2022). Business process re-engineering in public administration: The case study of Western Ligurian Sea Port Authority. *Sustainable Futures*, *4*, 100065.

[6] Riyadi, A., Kovacs, M., Serdült, U., & Kryssanov, V. (2023, February). A Machine Learning Approach to Government Business Process Re-engineering. In *2023 IEEE International Conference on Big Data and Smart Computing (BigComp)* (pp. 56-63). IEEE.

[7] Boullauazan, Y., Sys, C., & Vanelslander, T. (2023). Developing and demonstrating a maturity model for smart ports. *Maritime Policy & Management*, *50*(4), 447-465.

[8] Galli, G., Patrone, C., Battilani, C., & Revetria, R. (2021). Simulation and business process management notation in support of business process re-engineering.

In *Transactions on Engineering Technologies: World Congress on Engineering and Computer Science 2019* (pp. 47-58). Springer Singapore.

[9] Zheng, J., Yang, L., Li, W., Fu, X., & Li, D. (2022). Priority analysis of port investment along the 21st-Century Maritime Silk Road region: the case of Southeast Asia. *Maritime Policy & Management*, *49*(8), 1116-1134.

[10] Idris, H. (2022). Southeast asian port development: policy and initiatives towards achieving 2030 agenda on sustainable development goals. *Akademika*, *92*(2), 129-142.

[11] Nguyen, P. N., Woo, S. H., Beresford, A., & Pettit, S. (2020). Competition, market concentration, and relative efficiency of major container ports in Southeast Asia. *Journal of Transport Geography*, *83*, 102653.

[12] Wahyuni, S., Taufik, A. A., & Hui, F. K. P. (2020). Exploring key variables of port competitiveness: evidence from Indonesian ports. *Competitiveness Review: An International Business Journal*, *30*(5), 529-553.

[13] Jeevan, J., Othman, M. R., Salleh, N. H. M., Ghani, N. M. A., Noralam, N. A., & Divine Caesar, L. (2021). An analysis on the triadic connection between seaports, inland terminals and hinterland market. *Australian Journal of Maritime & Ocean Affairs*, *13*(1), 23-42.

[14] Irannezhad, E. (2020). The architectural design requirements of a blockchain-based port community system. *Logistics*, *4*(4), 30.

[15] Song, R., Cui, W., Vanthienen, J., Huang, L., & Wang, Y. (2022). Business process redesign towards IoT-enabled context-awareness: the case of a Chinese bulk port. *Business Process Management Journal*, *28*(3), 656-683.

[16] Ahmad, R. W., Hasan, H., Jayaraman, R., Salah, K., & Omar, M. (2021). Blockchain applications and architectures for port operations and logistics management. *Research in Transportation Business & Management*, *41*, 100620.

[17] Serra, P., Fancello, G., Tonelli, R., & Marchesi, L. (2022). Application prospects of blockchain technology to support the development of interport communities. *Computers*, *11*(5), 60.

[18] Sangeerth, P. S., & Lakshmy, K. V. (2021, January). Blockchain based smart contracts in automation of shipping ports. In *2021 6th International Conference on Inventive Computation Technologies (ICICT)* (pp. 1248-1253). IEEE.

[19] Munim, Z. H., Dushenko, M., Jimenez, V. J., Shakil, M. H., & Imset, M. (2020). Big data and artificial intelligence in the maritime industry: a bibliometric review and future research directions. *Maritime Policy & Management*, *47*(5), 577-597.

[20] Yau, K. L. A., Peng, S., Qadir, J., Low, Y. C., & Ling, M. H. (2020). Towards smart port infrastructures: Enhancing port activities using information and communications technology. *Ieee Access*, *8*, 83387-83404.